

ASYMPTOTIC DISTRIBUTIONS OF QUADRATIC FUNCTIONALS OF PARTIAL SUMS

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SUMMARY

Let $X_1, X_2, \dots$ be i.i.d. r.v.'s and ψ a non-negative function on $(0,1)$; put $Z_n = n^{-1} \sum_{j=1}^{n-1} \psi_{jn} S_j^2$ where $S_j = X_1 + \dots + X_j$ and $\psi_{jn} = \int_{j/n}^{(j+1)/n} \psi(t) dt$. This paper gives conditions under which $D(Z_n - EZ_n) \rightarrow D(\int_0^1 \psi(t)[B(t)^2 - t] dt)$ with $B(t)$ standard Brownian motion; under the same conditions this asymptotic distribution is the distribution of $\sum \gamma_m (U_m^2 - 1)$ with $\{\gamma_m\}$ a positive real sequence and the U_m 's independent $N(0,1)$ r.v.'s. Analogous results hold for functionals of the form $Z_n^0 = n^{-1} \sum_{j=1}^{n-1} \psi_{jn}^0 (S_j - (j/n)S_n)^2$.

1. Introduction and main results

Let $X_1, X_2, \dots$ be independent and identically distributed random variables with $EX_i = 0$, $EX_i^2 = 1$ and $EX_i^4 = \mu_4 < \infty$. Put $S_0 = 0$ and $S_n = \sum_{i=1}^n X_i$ for $n = 1, 2, \dots$. Let $\psi(t)$ be a non-negative measurable function defined for $0 < t < 1$ and put $\psi_{in} = \int_{i/n}^{(i+1)/n} \psi(t) dt$. Finally, let

$$(1) \quad Z_n = n^{-1} \sum_{j=1}^{n-1} \psi_{jn} S_j^2$$

Then MacNeil ((1974), Theorem 1) proves that $D(Z_n) \rightarrow D(\int_0^1 \psi(t) B(t)^2 dt)$ under the assumption $\int_0^1 t \psi(t) dt < \infty$; here $B(t)$ denotes the standard Brownian motion process and $D(Z_n) \rightarrow$ refers to convergence in distribution of Z_n . MacNeill does not need the assumption $EX_i^4 < \infty$. Under this assumption a stronger result is proved below. Let

$$(2) \quad k(s, t) = \min(s, t) \psi(s)^{1/2} \psi(t)^{1/2} \quad \text{for } 0 < s, t < 1$$

Then we have

Theorem 1: If

$$(3) \quad \int_0^1 \int_0^1 k(s,t)^2 ds dt < \infty$$

then

$$(4) \quad D(Z_n - EZ_n) \rightarrow D\left(\int_0^1 \psi(t) \{B(t)^2 - t\} dt\right)$$

It is easy to show that $\int_0^1 t\psi(t)dt < \infty$ implies (3). An analogous improvement of Theorem 2 of MacNeill is contained in our next result. Let $\psi_{in}^0 = \int_{(2i-1)/2n}^{(2i+1)/2n} \psi(t)dt$, $i=1,2,\dots,n-1$ and put

$$(5) \quad Z_n^0 = n^{-1} \sum_{j=1}^{n-1} \psi_{jn}^0 (S_j - (j/n)S_n)^2$$

and

$$(6) \quad k^0(s,t) = (\min(s,t) - st)\psi(s)^{1/2}\psi(t)^{1/2}.$$

Theorem 2: If

$$(7) \quad \int_0^1 \int_0^1 k^0(s,t)^2 ds dt < \infty$$

then

$$(8) \quad D(Z_n^0 - EZ_n^0) \rightarrow D\left(\int_0^1 \psi(t) \{B^0(t)^2 - t(1-t)\} dt\right)$$

with $B^0(t)$ the standard Brownian bridge process.

Characterizations of the asymptotic distributions appearing in (4) and (8) are contained in Theorem 5 of MacNeill and Theorem 4.1 of Anderson and Darling (1952). Our next results improve on these. Let

$$(9) \quad c(u,v) = \int_{\max(u,v)}^1 \psi(t)dt \quad 0 < u,v < 1.$$

By change of the order of integration one finds that

$$(10) \quad \int_0^1 \int_0^1 c(u,v)^2 du dv = \int_0^1 \int_0^1 k(s,t)^2 ds dt.$$

Hence under (3) we can associate with the symmetric kernels c and k self-adjoint

Hilbert-Schmidt integral operators T_c and T_k on $L_2(0,1)$ and these are related (Pietsch (1963)) in the sense that there exist mappings $P: L_2(0,1) \rightarrow L_2(0,1)$ and $Q: L_2(0,1) \rightarrow L_2(0,1)$ such that $T_c = QP$ and $T_k = PQ$ (for $f \in L_2(0,1)$ take $(Pf)(x) = \int_0^1 \psi(x)^{1/2} I_{(0,x]}(y) f(y) dy$ and $(Qf)(x) = \int_0^1 \psi(y)^{1/2} I_{[x,1]}(y) f(y) dy$ with I_A the indicator function of the set A). It follows that T_c and T_k have the same positive eigenvalues $\{\gamma_m\}$ with $\sum_m \gamma_m^2$ equal to the integrals in (10). We have

Theorem 3:

Under (2)

$$(11) \quad D(\int_0^1 \psi(t) \{B(t)^2 - t\} dt) = D(\sum_{m=1}^{\infty} \gamma_m (U_m^2 - 1))$$

with $U_1, U_2, \dots$, independent $N(0,1)$ random variables.

Analogously, with

$$(12) \quad c^0(u,v) = \int_0^1 \chi(u,z) \chi(v,z) \psi(z) dz$$

where

$$(13) \quad \begin{aligned} \chi(u,z) &= 1 - z & \text{for } u \leq z \\ &= -z & \text{for } u > z \end{aligned}$$

we get

$$(14) \quad \int_0^1 \int_0^1 c^0(u,v)^2 du dv = \int_0^1 \int_0^1 k^0(s,t)^2 ds dt$$

and when this is finite a set of eigenvalues $\{\gamma_m^0\}$ can be associated with these two kernels just as in the previous case. We then have

Theorem 4:

Under (7)

$$(15) \quad D(\int_0^1 \psi(t) \{B^0(t)^2 - t(1-t)\} dt) = D(\sum_{m=1}^{\infty} \gamma_m^0 (U_m^2 - 1))$$

with $U_1, U_2, \dots$ as above.

We prove these results in the next paragraph and give further discussion in paragraph 3.

2. Proof of main results

Proof of Theorem 1:

First note that $E(B(t)^2 - t) = 0$ and $\text{cov}(B(s)^2 - s, B(t)^2 - t) = 2 \min(s,t)^2$; hence

$\text{cov}(\psi(s)(B(s)^2 - s), \psi(t)(B(t)^2 - t)) = 2k(s,t)^2$. Thus, under (2) the stochastic integral appearing in (4) exists in the quadratic mean sense (Loève (1960), p. 472) and (4) is to be interpreted in this sense. Let $Y_n(t) = n^{-1/2} S_{[nt]}$ where $[x]$ is the largest integer in x .

We apply Theorem 4.2 of Billingsley (1968, p. 25).

$$(16) \quad V_{nm} = \sum_{j=1}^{n-1} \psi_{jn}(Y_m(j/n)^2 - EY_m(j/n)^2)$$

and

$$(17) \quad V_n = \sum_{j=1}^{n-1} \psi_{jn}(B(j/n)^2 - j/n)$$

Since $EY_m(t)^2 = [mt]/m \rightarrow t$ as $m \rightarrow \infty$ and the joint finite dimensional distributions of $Y_m(t)$ converge to those of $B(t)$, it follows that $D(V_{nm}) \rightarrow D(V_n)$ as $m \rightarrow \infty$. Next we show that $D(V_n) \rightarrow D(\int_0^1 \psi(t) \{B(t)^2 - t\} dt)$ as $n \rightarrow \infty$. To this effect, let

$$(18) \quad W_n = \int_{1/n}^1 \psi(t) \{B(t)^2 - t\} dt.$$

Then clearly $D(W_n) \rightarrow D(\int_0^1 \psi(t) \{B(t)^2 - t\} dt)$; hence it suffices to show that

$$(19) \quad E(V_n - W_n)^2 \rightarrow 0.$$

Now

$$V_n - W_n = \sum_{j=1}^{n-1} \int_{j/n}^{(j+1)/n} \psi(t) \{B(j/n)^2 - j/n - (B(t)^2 - t)\} dt$$

so that

$$E(V_n - W_n)^2 = \sum_j \sum_k \int_{j/n}^{(j+1)/n} \int_{k/n}^{(k+1)/n} \psi(s) \psi(t) EA_{jkn}(s,t) ds dt$$

with

$$A_{jkn}(s,t) = \{B(j/n)^2 - j/n - (B(s)^2 - s)\} \{B(k/n)^2 - k/n - (B(t)^2 - t)\}.$$

We find that

$$EA_{jkn}(s,t) = 2 \{ \min(j/n, k/n)^2 - \min(s, k/n)^2 - \min(j/n, t)^2 + \min(s, t)^2 \}.$$

Thus, for $j \neq k$ we have $EA_{jkn}(s,t) = 0$ for $j/n \leq s \leq (j+1)/n$ and $k/n \leq t \leq (k+1)/n$; for $j = k$ we find $EA_{jjn}(s,t) = 2 \{ \min(s, t)^2 - (j/n)^2 \}$. Hence

$$\begin{aligned} E(V_n - W_n)^2 &= 2 \sum_{j=1}^{n-1} \int_{j/n}^{(j+1)/n} \int_{j/n}^{(j+1)/n} \psi(s) \psi(t) \{ \min(s, t)^2 - (j/n)^2 \} ds dt \\ &\leq 2 \int_{1/n}^1 \int_{1/n}^1 k(s,t)^2 ds dt \end{aligned}$$

with $I_n = \bigcup_{j=1}^{n-1} \{(s,t) : j/n \leq s, t \leq (j+1)/n\}$. By the dominated convergence theorem this bound tends to zero and (19) follows.

To complete the proof it remains to show that $\lim_{m \rightarrow \infty} \sup P(|V_{nm} - (Z_m - EZ_m)| \geq \epsilon) \rightarrow 0$ as $n \rightarrow \infty$ for all $\epsilon > 0$. It suffices to show that $\lim_{m \rightarrow \infty} \sup E(V_{nm} - (Z_m - EZ_m))^2 \rightarrow 0$ as $n \rightarrow \infty$. Now put

$$T_m = \int_{1/m}^1 \psi(t) (Y_m(t)^2 - EY_m(t)^2) dt.$$

Then it suffices to show that $\lim_{m \rightarrow \infty} \sup E(V_{nm} - T_m)^2 \rightarrow 0$ as $n \rightarrow \infty$ and $E(T_m - (Z_m - EZ_m))^2 \rightarrow 0$ as $m \rightarrow \infty$. For $m > n$

$$(20) \quad V_{nm} - T_m = \sum_{j=1}^{n-1} \int_{j/n}^{(j+1)/n} \psi(t) \{Y_m(j/n)^2 - EY_m(j/n)^2\} \\ - \{Y_m(t)^2 - EY_m(t)^2\} dt - \int_{1/m}^1 \psi(t) \{Y_m(t)^2 - EY_m(t)^2\} dt.$$

The square of the sum on the right-hand side has expectation

$$(21) \quad \sum_j \sum_k \int_{j/n}^{(j+1)/n} \int_{k/n}^{(k+1)/n} \psi(s) \psi(t) EC_{jknm}(s,t) ds dt$$

with

$$C_{jknm}(s,t) = \{Y_m(j/n)^2 - EY_m(j/n)^2\} - \{Y_m(s)^2 - EY_m(s)^2\} \\ \{Y_m(k/n)^2 - EY_m(k/n)^2\} - \{Y_m(t)^2 - EY_m(t)^2\}.$$

Now from its definition we get

$$\text{cov}(Y_m(s)^2, Y_m(t)^2) \\ = m^{-2} \{(\mu_4 - 3) \min([ms], [mt]) + 2 \min([ms], [mt])^2\} \leq (\mu_4 + 5) \min(s,t)^2$$

From this expression it follows that $EC_{jknm}(s,t) = 0$ for $j \neq k$ and $j/n \leq s \leq (j+1)/n$, $k/n \leq t \leq (k+1)/n$. Moreover for $m > n$ we have $EC_{jjnm} \leq (\mu_4 + 5) \min(s,t)^2$ for $j/n \leq s, t \leq (j+1)/n$. Hence (21) is bounded by $(\mu_4 + 5) \int_{I_n} \int_{I_n} k(s,t)^2 ds dt$ with I_n as before. By the dominated convergence theorem this last bound goes to zero as $n \rightarrow \infty$.

The square of the second term on the right in (20) has expectation equal to

$$\int_{1/m}^1 \int_{1/m}^1 \psi(s) \psi(t) \text{cov}(Y_m(s)^2, Y_m(t)^2) ds dt \\ \leq (\mu_4 + 5) \int_{1/m}^1 \int_{1/m}^1 k(s,t)^2 ds dt$$

$\rightarrow (\mu_4 + 5) \int_0^{1/n} \int_0^{1/n} n_k(s,t)^2 ds dt$ as $m \rightarrow \infty$, as $n \rightarrow \infty$ this tends to zero.

Hence it remains to show that $E(T_m - (Z_m - EZ_m))^2 \rightarrow 0$ as $m \rightarrow \infty$. We have

$$E(T_m - (Z_m - EZ_m))^2 = \sum_{j,k=1}^{m-1} \int_{k/m}^{(k+1)/m} \int_{j/m}^{(j+1)/m} \psi(s)\psi(t) ED_{jkm}(s,t) ds dt$$

where

$$D_{jkm}(s,t) = \{(Y_m(s))^2 - EY_m(s)^2 - (Y_m(j/m))^2 - EY_m(j/m)^2\} \\ \{ (Y_m(t))^2 - EY_m(t)^2 - (Y_m(k/m))^2 - EY_m(k/m)^2 \}$$

From this it follows that for $j \neq k$ $ED_{jkm}(s,t)=0$ for $j/m < s \leq (j+1)/m$ and $k/m \leq t \leq (k+1)/m$, and $ED_{jjm}(s,t) \leq (\mu_4 + 5) \min(s,t)^2$ for $j/m \leq s, t \leq (j+1)/m$. Hence

$$E(T_m - (Z_m - EZ_m))^2 \leq (\mu_4 + 5) \int_{1/m}^1 \int_{1/m}^1 k(s,t)^2 ds dt.$$

As before this bound goes to zero as $m \rightarrow \infty$. This completes the proof.

Proof of Theorem 2:

This is entirely analogous to the above and the details are omitted.

Proof of Theorem 3:

For positive integers n define

$$(22) \quad \psi_n(t) = \psi(t) \quad \text{for } t \geq 1/n \\ = 0 \quad \text{for } t < 1/n$$

Then W_n as defined in (18) is given by

$$(23) \quad W_n = \int_0^1 \psi_n(t) \{B(t)^2 - t\} dt.$$

Since $\psi_n \in L_1(0,1)$ it follows from a slight extension of Theorem 3 of Sawyer (1972, p. 283) that

$$D(W_n) = D(\sum_{m=1}^{\infty} \gamma_{nm} (U_m^2 - 1))$$

with $\{\gamma_{nm}\}_{m=1,2,\dots}$ the eigenvalues of the kernel

$$c_n(u,v) = \int_{\max(u,v)}^1 \psi_n(t) dt.$$

We assume that for each n these eigenvalues are indexed such that $\gamma_{n1} \geq \gamma_{n2} \geq \dots$. Since $D(W_n) \rightarrow D(\int_0^1 \psi(t) \{B(t)^2 - t\} dt)$ as $n \rightarrow \infty$, it suffices to show that

$$D(\sum_m \gamma_{nm} (U_m^2 - 1)) \rightarrow D(\sum_m \gamma_m (U_m^2 - 1)) \text{ as } n \rightarrow \infty.$$

This will follow if we can show that

$$\sum_m (\gamma_{nm} - \gamma_m)^2 \rightarrow 0 \text{ as } n \rightarrow \infty.$$

Examination of the proof of lemma 5 of Dunford and Schwarz (1966, p. 1091) shows that $\gamma_{nm} \rightarrow \gamma_m$ as $n \rightarrow \infty$ for each m if the γ_m 's are indexed such that $\gamma_1 \geq \gamma_2 \geq \dots$. Then

$$\begin{aligned} & \sum_{m=1}^{\infty} (\gamma_{nm} - \gamma_m)^2 \\ &= \sum_{m=1}^M (\gamma_{nm} - \gamma_m)^2 + \sum_{m=M+1}^{\infty} (\gamma_{nm} - \gamma_m)^2 \\ &\leq \sum_{m=1}^M (\gamma_{nm} - \gamma_m)^2 + 2 \sum_{m=M+1}^{\infty} \gamma_{nm}^2 + 2 \sum_{m=M+1}^{\infty} \gamma_m^2 \\ &= \sum_{m=1}^M (\gamma_{nm} - \gamma_m)^2 + 2(\sum_{m=1}^{\infty} \gamma_{nm}^2 - \sum_{m=1}^M \gamma_{nm}^2) + 2 \sum_{m=M+1}^{\infty} \gamma_m^2 \end{aligned}$$

Now

$$\sum_{m=1}^{\infty} \gamma_{nm}^2 = \int_0^1 \int_0^1 c_n(u,v)^2 du dv \rightarrow \int_0^1 \int_0^1 c(u,v)^2 du dv = \sum_{m=1}^{\infty} \gamma_m^2.$$

Hence we find

$$\begin{aligned} \limsup_{n \rightarrow \infty} \sum_{m=1}^{\infty} (\gamma_{nm} - \gamma_m)^2 &\leq 2(\sum_{m=1}^{\infty} \gamma_m^2 - \sum_{m=1}^M \gamma_m^2) + 2 \sum_{m=M+1}^{\infty} \gamma_m^2 \\ &= 4 \sum_{m=M+1}^{\infty} \gamma_m^2 \end{aligned}$$

By taking M large enough this bound becomes arbitrarily small; hence the result follows.

Proof of Theorem 4:

This is again analogous to the previous and the details are omitted.

3. Discussion

- 3.1 When computing the eigenvalues appearing above one has the choice of using either of the kernels k or c (resp. k^0 or c^0). MacNeill (resp. Anderson and Darling) proceeds from k (resp. k^0) and shows that the problem reduces to a Sturm-Liouville boundary value problem. It is easy to see that calculation of the eigenvalues γ and corresponding eigenfunctions g of the kernel c amounts to another Sturm-Liouville problem, viz. solve

$$\gamma d(g'(x)/\psi(x))/dx + g(x) = 0$$

subject to $g(x) \rightarrow 0$ as $x \uparrow 1$ and $g'(x)/\psi(x) \rightarrow 0$ as $x \downarrow 0$. For example, with $\psi(x) = x^k$ and $k > -2$, this is solved by $g(x) = g_j(x) = x^{(k+2)/2} J_{-(k+1)/(k+2)}(\gamma_j x^{(k+2)/2})$, $\gamma_j = 4/(k+2)^2 \alpha_j^2$ with $J(\nu; x)$ a Bessel function of the first kind of order ν and α_j its j -th zero. This agrees with MacNeill's result for this case.

The eigenvalues of c^0 for a variety of choices of ψ are given in De Wet and Venter (1973). There W is the notation used for the present ψ .

- 3.2 Theorems 2 and 4 have application to tests of fit closely related to those discussed by De Wet and Venter (1973). If we replace the weights $W(k/(n+1))$ used in equation (2) of De Wet and Venter by

$$(n+1)\psi_{kn+1} = (n+1) \int_{(2k-1)/(2n+2)}^{(2k+1)/(2n+2)} \psi(t) dt, \quad k=1,2,\dots,n$$

and call the resulting statistics Q_n^ψ , then it is easy to see that under H_0 we have $Q_n^\psi = ((n+1)/S_{n+1}^*)^2 Z_{n+1}^0$ where $(n+1)/S_{n+1}^* \rightarrow 1$ a.s. and Z_{n+1}^0 is as defined by (5) above. The asymptotic null distribution of $Q_n^\psi - EQ_n^\psi$ will therefore generally coincide with that of $Z_n^0 - EZ_n^0$ and be obtainable from Theorems 2 and 4 above. For example, when $\psi(u) = 1/\phi(\Phi^{-1}(u))^2$ with ϕ and Φ the standard normal density and distribution functions respectively, it is quite easy to apply the present results whereas a great deal of work was required with the approach of De Wet and Venter (1973).

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